

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
KHOA CÔNG NGHỆ THÔNG TIN



BÁO CÁO ĐỀ CƯƠNG THỰC TẬP CƠ SỞ

ĐỀ TÀI:

**Nghiên cứu học đa nhiệm và cơ chế Attention trong bài toán
phân đoạn và phân loại u não từ ảnh MRI**

Giảng viên hướng dẫn: TS.Kim Ngọc Bách

Sinh viên thực hiện: Đặng Tuấn Minh

Mã Sinh Viên: B23DCCN539

HÀ NỘI, THÁNG 4/2026

Báo cáo tuần 8

I) Công việc hoàn thành

Sau khi giảm độ khắt khe của Data Augmentation từ tuần trước em đã train lại 4 model.

1.1) Classification

Running on cuda

HYPERPARAMETERS SUMMARY

LR_CLF : 0.0001
LR_SEG : 0.001
BATCH_CLF : 32
BATCH_SEG : 12
BATCH_JOINT : 12
EPOCHS : 40
OPTIMIZER : AdamW
LOSS_CLF : CrossEntropyLoss
LOSS_SEG : BCEWithLogitsLoss (Weighted) + DiceLoss
EARLY_STOP_PATIENCE : 10
WEIGHT_DECAY : 0.001

CLASSIFICATION TRAINING

Epoch 1: Train Acc=0.5098 | Val Acc=0.3410
Saved best model (Val Acc: 0.3410)
Epoch 2: Train Acc=0.6082 | Val Acc=0.6470
Saved best model (Val Acc: 0.6470)
Epoch 3: Train Acc=0.6468 | Val Acc=0.6740
Saved best model (Val Acc: 0.6740)
Epoch 4: Train Acc=0.6923 | Val Acc=0.5520
Epoch 5: Train Acc=0.7425 | Val Acc=0.7180
Saved best model (Val Acc: 0.7180)
Epoch 6: Train Acc=0.7810 | Val Acc=0.6600
Epoch 7: Train Acc=0.7907 | Val Acc=0.7580
Saved best model (Val Acc: 0.7580)
Epoch 8: Train Acc=0.8075 | Val Acc=0.7670
Saved best model (Val Acc: 0.7670)
Epoch 9: Train Acc=0.8253 | Val Acc=0.7020
Epoch 10: Train Acc=0.8393 | Val Acc=0.8520
Saved best model (Val Acc: 0.8520)
Epoch 11: Train Acc=0.8495 | Val Acc=0.7640
Epoch 12: Train Acc=0.8648 | Val Acc=0.8600
Saved best model (Val Acc: 0.8600)
Epoch 13: Train Acc=0.8648 | Val Acc=0.7830
Epoch 14: Train Acc=0.8798 | Val Acc=0.7150
Epoch 15: Train Acc=0.8970 | Val Acc=0.8740
Saved best model (Val Acc: 0.8740)
Epoch 16: Train Acc=0.8935 | Val Acc=0.8120
Epoch 17: Train Acc=0.9018 | Val Acc=0.9050

Saved best model (Val Acc: 0.9050)

Epoch 18: Train Acc=0.9055 | Val Acc=0.8300

Epoch 19: Train Acc=0.9180 | Val Acc=0.6880

Epoch 20: Train Acc=0.9140 | Val Acc=0.8620

Epoch 21: Train Acc=0.9215 | Val Acc=0.8060

Epoch 22: Train Acc=0.9203 | Val Acc=0.8460

Epoch 23: Train Acc=0.9183 | Val Acc=0.7990

Epoch 24: Train Acc=0.9237 | Val Acc=0.8970

Epoch 25: Train Acc=0.9200 | Val Acc=0.8550

Epoch 26: Train Acc=0.9315 | Val Acc=0.8980

Epoch 27: Train Acc=0.9410 | Val Acc=0.8790

Early stopping triggered at epoch 27

EVALUATING CLASSIFICATION ON TEST SET...

Test Acc: 0.9050 | Prec: 0.9080 | Recall: 0.9050 | F1: 0.9025

1.2 Segmentation Using Base Unet

Running on cuda

HYPERPARAMETERS SUMMARY

LR_CLF : 0.0001

LR_SEG : 0.001

BATCH_CLF : 32

BATCH_SEG : 12

BATCH_JOINT : 12

EPOCHS : 40

OPTIMIZER : AdamW

LOSS_CLF : CrossEntropyLoss

LOSS_SEG : BCEWithLogitsLoss (Weighted) + DiceLoss

EARLY_STOP_PATIENCE : 10

WEIGHT_DECAY : 0.001

SEGMENTATION TRAINING

Epoch 1: Val mIoU=0.0693 | Dice=0.1284 | Pixel Acc=0.8586

Saved best model (Val mIoU: 0.0693)

Epoch 2: Val mIoU=0.2417 | Dice=0.3841 | Pixel Acc=0.9797

Saved best model (Val mIoU: 0.2417)

Epoch 3: Val mIoU=0.3420 | Dice=0.5037 | Pixel Acc=0.9847

Saved best model (Val mIoU: 0.3420)

Epoch 4: Val mIoU=0.4211 | Dice=0.5872 | Pixel Acc=0.9877

Saved best model (Val mIoU: 0.4211)

Epoch 5: Val mIoU=0.4307 | Dice=0.5970 | Pixel Acc=0.9865

Saved best model (Val mIoU: 0.4307)

Epoch 6: Val mIoU=0.5045 | Dice=0.6654 | Pixel Acc=0.9895

Saved best model (Val mIoU: 0.5045)

Epoch 7: Val mIoU=0.4894 | Dice=0.6529 | Pixel Acc=0.9877

Epoch 8: Val mIoU=0.5234 | Dice=0.6806 | Pixel Acc=0.9906
Saved best model (Val mIoU: 0.5234)
Epoch 9: Val mIoU=0.4925 | Dice=0.6539 | Pixel Acc=0.9909
Epoch 10: Val mIoU=0.5562 | Dice=0.7104 | Pixel Acc=0.9905
Saved best model (Val mIoU: 0.5562)
Epoch 11: Val mIoU=0.5968 | Dice=0.7429 | Pixel Acc=0.9917
Saved best model (Val mIoU: 0.5968)
Epoch 12: Val mIoU=0.5925 | Dice=0.7406 | Pixel Acc=0.9908
Epoch 13: Val mIoU=0.6192 | Dice=0.7607 | Pixel Acc=0.9925
Saved best model (Val mIoU: 0.6192)
Epoch 14: Val mIoU=0.6015 | Dice=0.7463 | Pixel Acc=0.9918
Epoch 15: Val mIoU=0.5816 | Dice=0.7306 | Pixel Acc=0.9906
Epoch 16: Val mIoU=0.6295 | Dice=0.7674 | Pixel Acc=0.9925
Saved best model (Val mIoU: 0.6295)
Epoch 17: Val mIoU=0.6039 | Dice=0.7478 | Pixel Acc=0.9922
Epoch 18: Val mIoU=0.6326 | Dice=0.7710 | Pixel Acc=0.9924
Saved best model (Val mIoU: 0.6326)
Epoch 19: Val mIoU=0.5684 | Dice=0.7197 | Pixel Acc=0.9893
Epoch 20: Val mIoU=0.6478 | Dice=0.7823 | Pixel Acc=0.9928
Saved best model (Val mIoU: 0.6478)
Epoch 21: Val mIoU=0.6521 | Dice=0.7857 | Pixel Acc=0.9933
Saved best model (Val mIoU: 0.6521)
Epoch 22: Val mIoU=0.5833 | Dice=0.7311 | Pixel Acc=0.9908
Epoch 23: Val mIoU=0.6673 | Dice=0.7973 | Pixel Acc=0.9934
Saved best model (Val mIoU: 0.6673)
Epoch 24: Val mIoU=0.6433 | Dice=0.7789 | Pixel Acc=0.9933
Epoch 25: Val mIoU=0.6422 | Dice=0.7780 | Pixel Acc=0.9930
Epoch 26: Val mIoU=0.6042 | Dice=0.7495 | Pixel Acc=0.9906
Epoch 27: Val mIoU=0.6593 | Dice=0.7905 | Pixel Acc=0.9931
Epoch 28: Val mIoU=0.6527 | Dice=0.7859 | Pixel Acc=0.9932
Epoch 29: Val mIoU=0.6245 | Dice=0.7652 | Pixel Acc=0.9916
Epoch 30: Val mIoU=0.6913 | Dice=0.8143 | Pixel Acc=0.9941
Saved best model (Val mIoU: 0.6913)
Epoch 31: Val mIoU=0.6764 | Dice=0.8030 | Pixel Acc=0.9937
Epoch 32: Val mIoU=0.6833 | Dice=0.8090 | Pixel Acc=0.9936
Epoch 33: Val mIoU=0.6883 | Dice=0.8123 | Pixel Acc=0.9938
Epoch 34: Val mIoU=0.6918 | Dice=0.8148 | Pixel Acc=0.9938
Saved best model (Val mIoU: 0.6918)
Epoch 35: Val mIoU=0.6918 | Dice=0.8142 | Pixel Acc=0.9939
Saved best model (Val mIoU: 0.6918)
Epoch 36: Val mIoU=0.6965 | Dice=0.8178 | Pixel Acc=0.9940
Saved best model (Val mIoU: 0.6965)
Epoch 37: Val mIoU=0.6953 | Dice=0.8171 | Pixel Acc=0.9942
Epoch 38: Val mIoU=0.6933 | Dice=0.8152 | Pixel Acc=0.9940

Epoch 39: Val mIoU=0.6800 | Dice=0.8054 | Pixel Acc=0.9938
Epoch 40: Val mIoU=0.6689 | Dice=0.7979 | Pixel Acc=0.9930

EVALUATING UNET-SEGMENTATION ON TEST SET...

Test mIoU: 0.7409 | Dice: 0.8445 | Pixel Acc: 0.9948

1.3 Multitask Learning

Running on cuda

HYPERPARAMETERS SUMMARY

LR_CLF : 0.0001
LR_SEG : 0.001
BATCH_CLF : 32
BATCH_SEG : 12
BATCH_JOINT : 12
EPOCHS : 40
OPTIMIZER : AdamW
LOSS_CLF : CrossEntropyLoss
LOSS_SEG : BCEWithLogitsLoss (Weighted) + DiceLoss
EARLY_STOP_PATIENCE : 10
WEIGHT_DECAY : 0.001

JOINT TRAINING

Epoch 1: Clf Acc=0.6089 | mIoU=0.0888 | Dice=0.1610 | PixAcc=0.9120
Saved best model (Combined Score: 0.3489)
Epoch 2: Clf Acc=0.5427 | mIoU=0.1345 | Dice=0.2340 | PixAcc=0.9758
Epoch 3: Clf Acc=0.4977 | mIoU=0.1930 | Dice=0.3167 | PixAcc=0.9831
Epoch 4: Clf Acc=0.6746 | mIoU=0.2795 | Dice=0.4317 | PixAcc=0.9786
Saved best model (Combined Score: 0.4770)
Epoch 5: Clf Acc=0.6430 | mIoU=0.3407 | Dice=0.5023 | PixAcc=0.9853
Saved best model (Combined Score: 0.4919)
Epoch 6: Clf Acc=0.5987 | mIoU=0.3287 | Dice=0.4879 | PixAcc=0.9813
Epoch 7: Clf Acc=0.7554 | mIoU=0.4276 | Dice=0.5939 | PixAcc=0.9869
Saved best model (Combined Score: 0.5915)
Epoch 8: Clf Acc=0.6405 | mIoU=0.3632 | Dice=0.5261 | PixAcc=0.9809
Epoch 9: Clf Acc=0.7680 | mIoU=0.4845 | Dice=0.6476 | PixAcc=0.9894
Saved best model (Combined Score: 0.6263)
Epoch 10: Clf Acc=0.7760 | mIoU=0.4868 | Dice=0.6481 | PixAcc=0.9886
Saved best model (Combined Score: 0.6314)
Epoch 11: Clf Acc=0.7664 | mIoU=0.4863 | Dice=0.6469 | PixAcc=0.9900
Epoch 12: Clf Acc=0.8460 | mIoU=0.4503 | Dice=0.6161 | PixAcc=0.9851
Saved best model (Combined Score: 0.6481)
Epoch 13: Clf Acc=0.8337 | mIoU=0.5152 | Dice=0.6742 | PixAcc=0.9883
Saved best model (Combined Score: 0.6744)
Epoch 14: Clf Acc=0.8662 | mIoU=0.5125 | Dice=0.6744 | PixAcc=0.9881
Saved best model (Combined Score: 0.6893)

Epoch 15: Clf Acc=0.8864 | mIoU=0.5713 | Dice=0.7220 | PixAcc=0.9915
Saved best model (Combined Score: 0.7289)
Epoch 16: Clf Acc=0.8763 | mIoU=0.5993 | Dice=0.7449 | PixAcc=0.9924
Saved best model (Combined Score: 0.7378)
Epoch 17: Clf Acc=0.8829 | mIoU=0.5458 | Dice=0.7023 | PixAcc=0.9889
Epoch 18: Clf Acc=0.8791 | mIoU=0.5360 | Dice=0.6901 | PixAcc=0.9917
Epoch 19: Clf Acc=0.8842 | mIoU=0.6145 | Dice=0.7567 | PixAcc=0.9924
Saved best model (Combined Score: 0.7494)
Epoch 20: Clf Acc=0.7781 | mIoU=0.6003 | Dice=0.7469 | PixAcc=0.9913
Epoch 21: Clf Acc=0.9091 | mIoU=0.6302 | Dice=0.7700 | PixAcc=0.9927
Saved best model (Combined Score: 0.7696)
Epoch 22: Clf Acc=0.9104 | mIoU=0.6505 | Dice=0.7842 | PixAcc=0.9933
Saved best model (Combined Score: 0.7804)
Epoch 23: Clf Acc=0.7877 | mIoU=0.5898 | Dice=0.7373 | PixAcc=0.9921
Epoch 24: Clf Acc=0.9255 | mIoU=0.5476 | Dice=0.7029 | PixAcc=0.9881
Epoch 25: Clf Acc=0.9369 | mIoU=0.6589 | Dice=0.7912 | PixAcc=0.9933
Saved best model (Combined Score: 0.7979)
Epoch 26: Clf Acc=0.9154 | mIoU=0.6207 | Dice=0.7635 | PixAcc=0.9915
Epoch 27: Clf Acc=0.9208 | mIoU=0.6317 | Dice=0.7704 | PixAcc=0.9920
Epoch 28: Clf Acc=0.9192 | mIoU=0.6344 | Dice=0.7732 | PixAcc=0.9920
Epoch 29: Clf Acc=0.9141 | mIoU=0.6598 | Dice=0.7908 | PixAcc=0.9936
Epoch 30: Clf Acc=0.9167 | mIoU=0.6616 | Dice=0.7922 | PixAcc=0.9934
Epoch 31: Clf Acc=0.9091 | mIoU=0.6649 | Dice=0.7954 | PixAcc=0.9933
Epoch 32: Clf Acc=0.9432 | mIoU=0.6827 | Dice=0.8080 | PixAcc=0.9938
Saved best model (Combined Score: 0.8129)
Epoch 33: Clf Acc=0.9520 | mIoU=0.6898 | Dice=0.8140 | PixAcc=0.9940
Saved best model (Combined Score: 0.8209)
Epoch 34: Clf Acc=0.9583 | mIoU=0.6827 | Dice=0.8086 | PixAcc=0.9938
Epoch 35: Clf Acc=0.9381 | mIoU=0.6802 | Dice=0.8066 | PixAcc=0.9937
Epoch 36: Clf Acc=0.9571 | mIoU=0.6754 | Dice=0.8029 | PixAcc=0.9935
Epoch 37: Clf Acc=0.9448 | mIoU=0.6593 | Dice=0.7914 | PixAcc=0.9930
Epoch 38: Clf Acc=0.9293 | mIoU=0.6842 | Dice=0.8095 | PixAcc=0.9939
Epoch 39: Clf Acc=0.9646 | mIoU=0.6753 | Dice=0.8033 | PixAcc=0.9935
Epoch 40: Clf Acc=0.9562 | mIoU=0.6982 | Dice=0.8187 | PixAcc=0.9943
Saved best model (Combined Score: 0.8272)

Generating Joint Model Confusion Matrix...

EVALUATING JOINT MODEL ON TEST SET...

Test Clf Acc: 0.9698 | mIoU: 0.7351 | Dice: 0.8398 | Pixel Acc: 0.9947

1.4 Segmentation Using Attention Unet

Running on cuda

HYPERPARAMETERS SUMMARY

LR_CLF : 0.0001
LR_SEG : 0.001
BATCH_CLF : 32
BATCH_SEG : 12
BATCH_JOINT : 12
EPOCHS : 40
OPTIMIZER : AdamW
LOSS_CLF : CrossEntropyLoss
LOSS_SEG : BCEWithLogitsLoss (Weighted) + DiceLoss
EARLY_STOP_PATIENCE : 10
WEIGHT_DECAY : 0.001

ATTENTION U-NET TRAINING

Epoch 1: Val mIoU=0.0440 | Dice=0.0840 | Pixel Acc=0.7308
Saved best model (Val mIoU: 0.0440)
Epoch 2: Val mIoU=0.2097 | Dice=0.3406 | Pixel Acc=0.9670
Saved best model (Val mIoU: 0.2097)
Epoch 3: Val mIoU=0.3567 | Dice=0.5202 | Pixel Acc=0.9832
Saved best model (Val mIoU: 0.3567)
Epoch 4: Val mIoU=0.4134 | Dice=0.5801 | Pixel Acc=0.9860
Saved best model (Val mIoU: 0.4134)
Epoch 5: Val mIoU=0.4612 | Dice=0.6251 | Pixel Acc=0.9892
Saved best model (Val mIoU: 0.4612)
Epoch 6: Val mIoU=0.5220 | Dice=0.6801 | Pixel Acc=0.9904
Saved best model (Val mIoU: 0.5220)
Epoch 7: Val mIoU=0.5210 | Dice=0.6802 | Pixel Acc=0.9906
Epoch 8: Val mIoU=0.5506 | Dice=0.7068 | Pixel Acc=0.9901
Saved best model (Val mIoU: 0.5506)
Epoch 9: Val mIoU=0.5591 | Dice=0.7132 | Pixel Acc=0.9904
Saved best model (Val mIoU: 0.5591)
Epoch 10: Val mIoU=0.5708 | Dice=0.7223 | Pixel Acc=0.9906
Saved best model (Val mIoU: 0.5708)
Epoch 11: Val mIoU=0.5597 | Dice=0.7133 | Pixel Acc=0.9901
Epoch 12: Val mIoU=0.5895 | Dice=0.7388 | Pixel Acc=0.9914
Saved best model (Val mIoU: 0.5895)
Epoch 13: Val mIoU=0.6008 | Dice=0.7461 | Pixel Acc=0.9916
Saved best model (Val mIoU: 0.6008)
Epoch 14: Val mIoU=0.6203 | Dice=0.7621 | Pixel Acc=0.9924
Saved best model (Val mIoU: 0.6203)
Epoch 15: Val mIoU=0.6291 | Dice=0.7684 | Pixel Acc=0.9927
Saved best model (Val mIoU: 0.6291)
Epoch 16: Val mIoU=0.6258 | Dice=0.7662 | Pixel Acc=0.9924

Epoch 17: Val mIoU=0.6303 | Dice=0.7691 | Pixel Acc=0.9924
Saved best model (Val mIoU: 0.6303)
Epoch 18: Val mIoU=0.6146 | Dice=0.7565 | Pixel Acc=0.9930
Epoch 19: Val mIoU=0.6374 | Dice=0.7752 | Pixel Acc=0.9923
Saved best model (Val mIoU: 0.6374)
Epoch 20: Val mIoU=0.6319 | Dice=0.7703 | Pixel Acc=0.9929
Epoch 21: Val mIoU=0.6594 | Dice=0.7913 | Pixel Acc=0.9933
Saved best model (Val mIoU: 0.6594)
Epoch 22: Val mIoU=0.6244 | Dice=0.7651 | Pixel Acc=0.9924
Epoch 23: Val mIoU=0.6536 | Dice=0.7862 | Pixel Acc=0.9934
Epoch 24: Val mIoU=0.6503 | Dice=0.7848 | Pixel Acc=0.9935
Epoch 25: Val mIoU=0.6480 | Dice=0.7823 | Pixel Acc=0.9929
Epoch 26: Val mIoU=0.6594 | Dice=0.7913 | Pixel Acc=0.9934
Epoch 27: Val mIoU=0.6568 | Dice=0.7894 | Pixel Acc=0.9928
Epoch 28: Val mIoU=0.6800 | Dice=0.8063 | Pixel Acc=0.9938
Saved best model (Val mIoU: 0.6800)
Epoch 29: Val mIoU=0.6687 | Dice=0.7976 | Pixel Acc=0.9932
Epoch 30: Val mIoU=0.6858 | Dice=0.8105 | Pixel Acc=0.9937
Saved best model (Val mIoU: 0.6858)
Epoch 31: Val mIoU=0.6754 | Dice=0.8027 | Pixel Acc=0.9936
Epoch 32: Val mIoU=0.6808 | Dice=0.8069 | Pixel Acc=0.9938
Epoch 33: Val mIoU=0.6845 | Dice=0.8094 | Pixel Acc=0.9939
Epoch 34: Val mIoU=0.6843 | Dice=0.8094 | Pixel Acc=0.9936
Epoch 35: Val mIoU=0.6818 | Dice=0.8069 | Pixel Acc=0.9934
Epoch 36: Val mIoU=0.6830 | Dice=0.8080 | Pixel Acc=0.9939
Epoch 37: Val mIoU=0.7001 | Dice=0.8207 | Pixel Acc=0.9942
Saved best model (Val mIoU: 0.7001)
Epoch 38: Val mIoU=0.6944 | Dice=0.8166 | Pixel Acc=0.9940
Epoch 39: Val mIoU=0.6956 | Dice=0.8167 | Pixel Acc=0.9941
Epoch 40: Val mIoU=0.6884 | Dice=0.8124 | Pixel Acc=0.9938

EVALUATING ATTENTION U-NET ON TEST SET...

Test mIoU: 0.7430 | Dice: 0.8446 | Pixel Acc: 0.9949